

Indian Institute of Technology, Madras
Parallel Scientific Computing - ID5130

Kanak Agarwal

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1 One Dimensional Wave Equation

The one-dimensional wave equation is given by,

$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0 \quad (1)$$

with $c = 1$ and the initial condition,

$$u(x, t = 0) = \begin{cases} \sin(4\pi x), & \text{for } 0 \leq x \leq 0.5 \\ 0, & \text{for } 0.5 < x \leq 2 \end{cases} \quad (2)$$

The exact solution to this equation is given by,

$$u(x, t) = u_0(x - ct) \quad (3)$$

The first order upwind scheme for is given by,

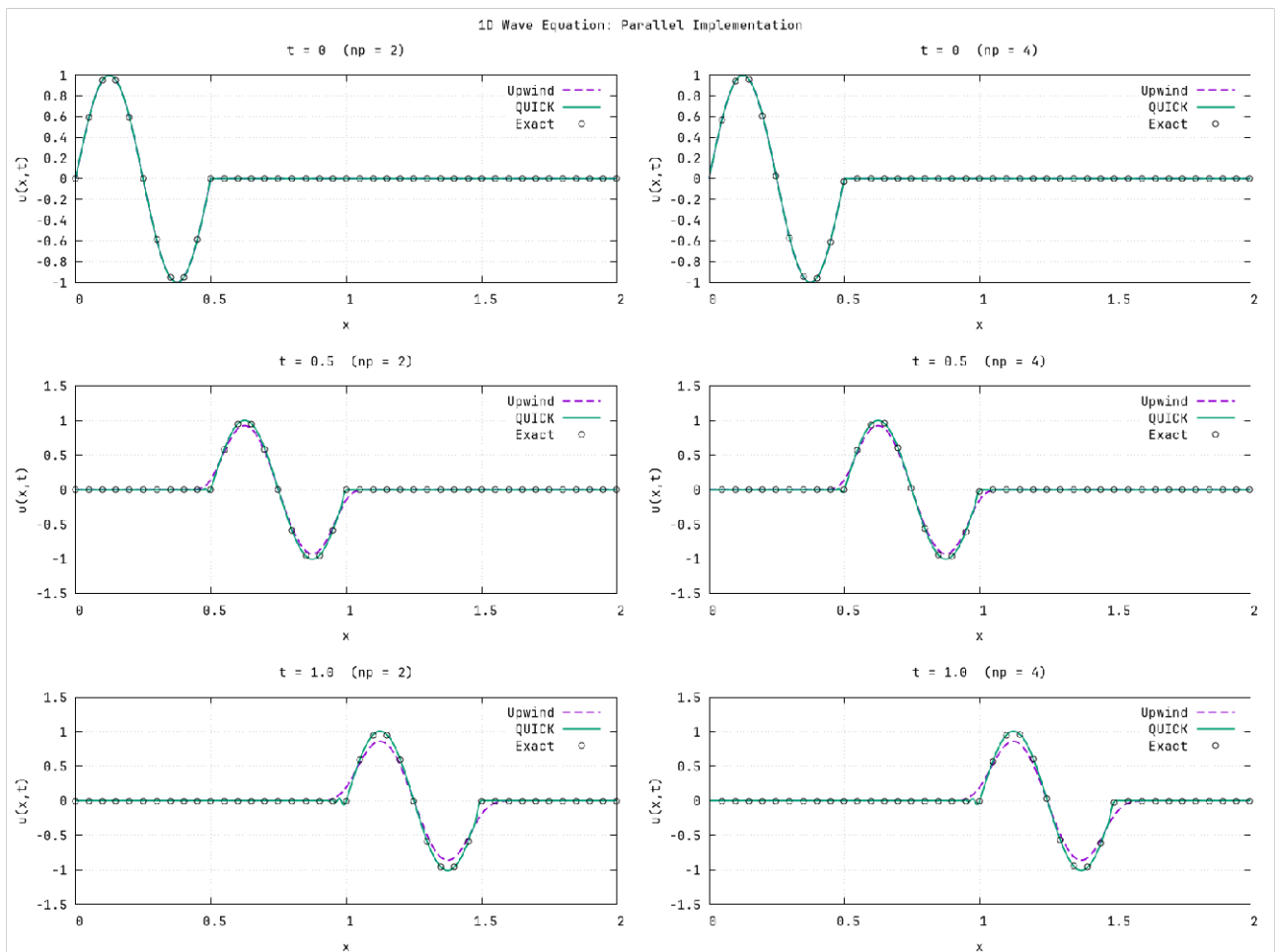
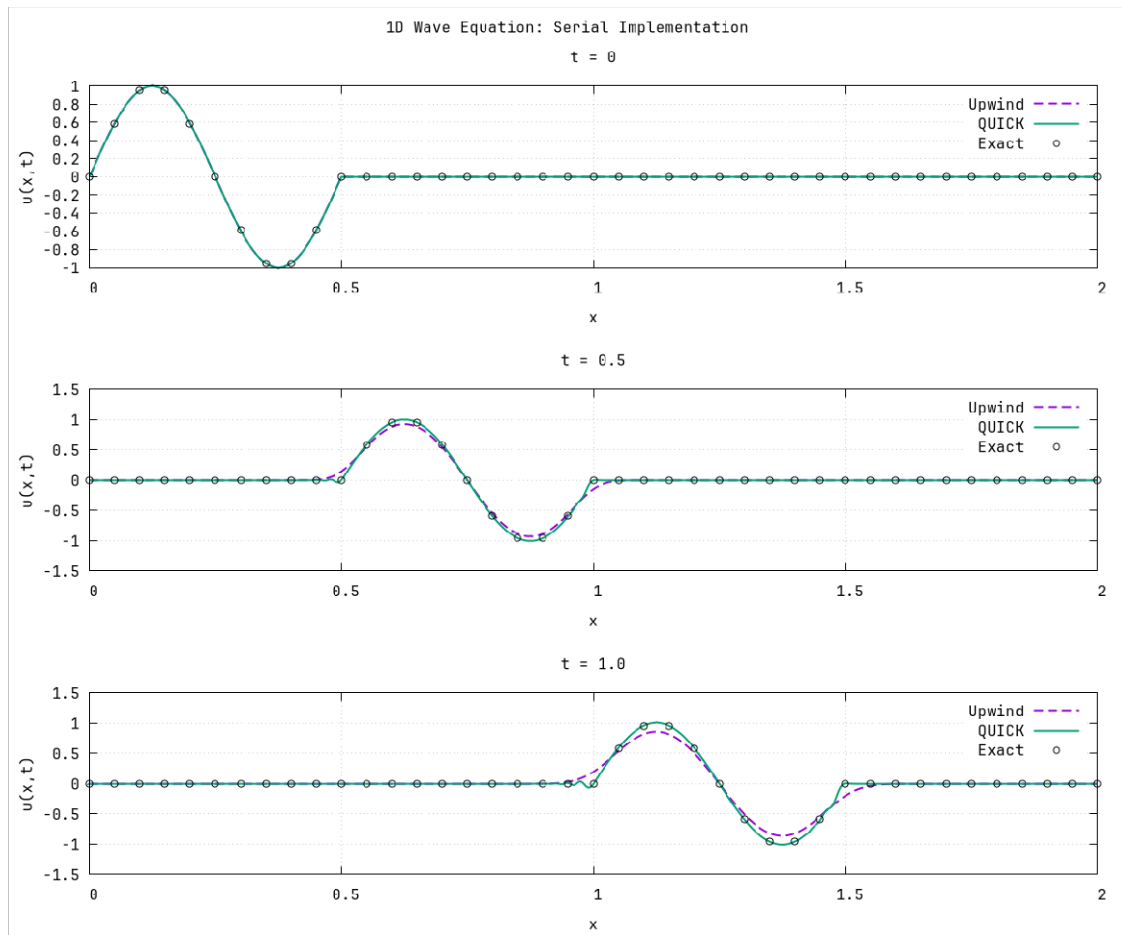
$$\frac{\partial u}{\partial x} \approx \frac{u_i^n - u_{i-1}^n}{\Delta x} \quad (4)$$

and the QUICK scheme is given by,

$$\frac{\partial u}{\partial x} \approx \frac{1}{\Delta x} \left(\frac{3}{8}u_i^n - \frac{7}{8}u_{i-1}^n + \frac{1}{8}u_{i-2}^n + \frac{3}{8}u_{i+1}^n \right) \quad (5)$$

where, $\Delta x = 0.002$ and $\Delta t = 0.0001$. The results for the serial and parallel implementations of the above scheme are shown below.

The upwind scheme is first order accurate and hence, it fails to accurately capture the peaks accurately and deteriorates with time. The QUICK scheme on the other hand, is third order accurate and hence, it captures the peaks much better than the upwind scheme. However, near the wavefront, the QUICK scheme shows some oscillations potentially due to the dispersive nature of the scheme. The results of the serial and parallel implementations are consistent.



2 Poisson Equation

The Poisson equation is given by,

$$\nabla^2 \phi = -q, \quad q = (x^2 + y^2) \quad (6)$$

The discretized Jacobi iteration for the Poisson equation is given by,

$$\phi_{i,j}^{k+1} = \frac{1}{4} (\phi_{i+1,j}^k + \phi_{i-1,j}^k + \phi_{i,j+1}^k + \phi_{i,j-1}^k) + \frac{\Delta^2}{4} q_{i,j} \quad (7)$$

The boundary conditions are as follows,

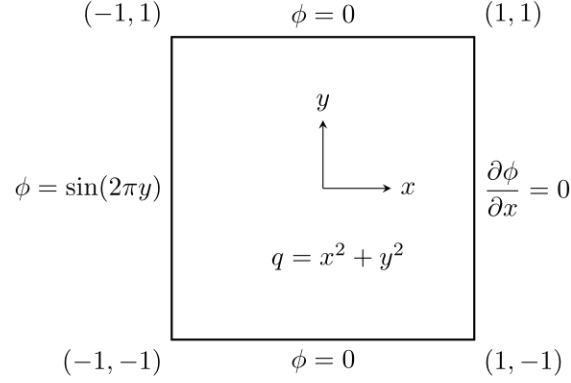


Figure 1: Domain and boundary conditions for the Poisson equation.

The results for the serial and parallel implementations of the above scheme are shown below. The lower resolution simulation ($\Delta x = 0.1$) converged in 386 iterations, while the higher resolution simulation ($\Delta x = 0.01$) converged in 1014 iterations in both the serial and parallel implementations. Convergence is determined using the ℓ_∞ norm and terminated when the residual falls below a specified tolerance (10^{-4}). The results of the serial and parallel implementations are consistent.

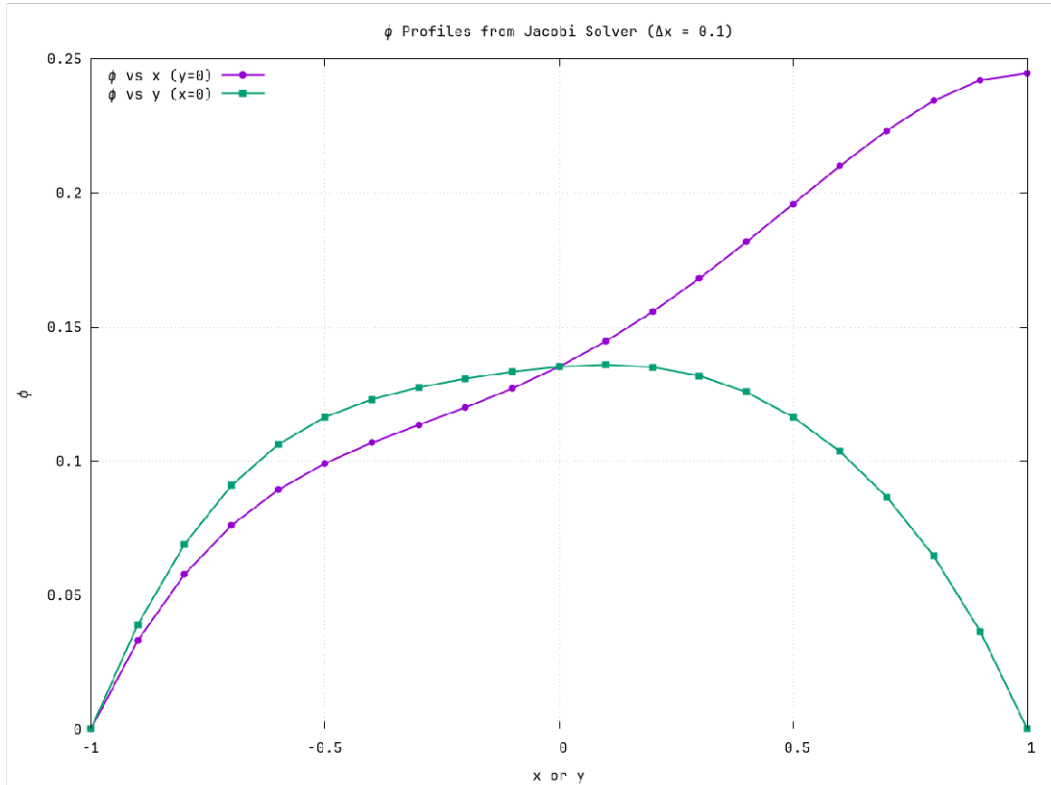


Figure 2: Serial implementation of the Poisson equation.

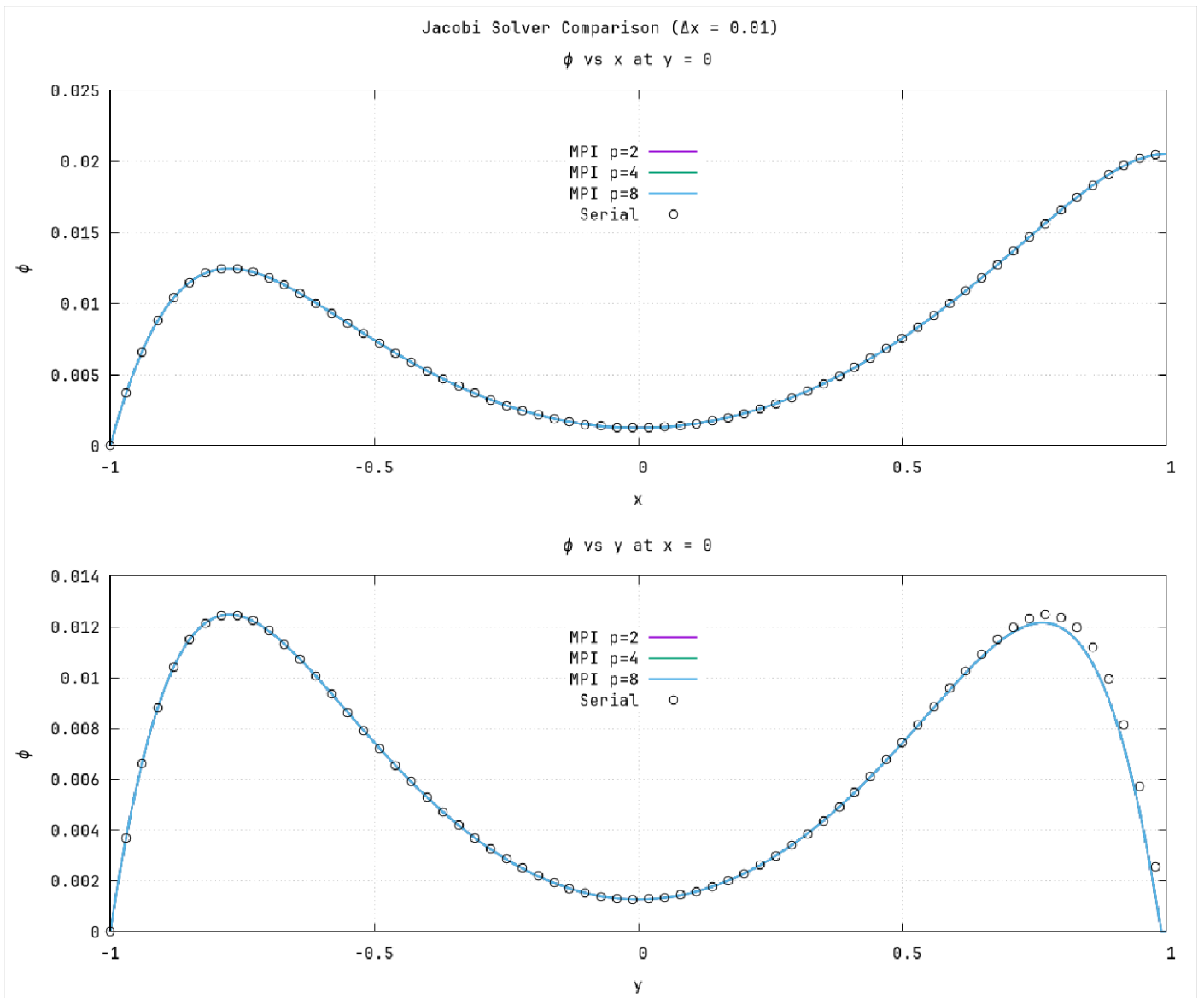


Figure 3: Parallel implementation of the Poisson equation.